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2. Diket : Enkripsi blok 3x3 citra logo (grayscale 0-255)

$E = A \cdot P$

$$A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix}$$

P = blok piksel

Dekripsi $\rightarrow$, $P = A^{-1} \cdot E$

Dit :

a. Tulis matriks A dgn parameter personalisasi. Hitung det(A) menggunakan ekspansi kofaktor baris pertama.

Jawab : $a = 1$, $b = 3$

$$A = \begin{bmatrix} 1+1 & 3 & 2 \\ 1 & 1+2 & 3 \\ 3 & 1 & 1+3 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 2 \\ 1 & 3 & 3 \\ 3 & 1 & 4 \end{bmatrix}$$

$$\det(A) = 2\begin{vmatrix} 3 & 3 \\ 1 & 4 \end{vmatrix} - 3\begin{vmatrix} 1 & 3 \\ 3 & 4 \end{vmatrix} + 2\begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix}$$

$$= 2((3 \cdot 4) - (3 \cdot 1)) - 3((4 \cdot 1) - (3 \cdot 3)) + 2((1 \cdot 1) - (3 \cdot 3))$$

$$= 2(12 - 3) - 3(4 - 9) + 2(1 - 9)$$

$$= 2(9) - 3(-5) + 2(-8)$$

$$= 18 + 15 - 16$$

$$= 17$$

b. Hitung semua matriks minor M_{ij} untuk semua $i, j \in \{1, 2, 3\}$.

Jawab :

$$M_{11} = \begin{vmatrix} 3 & 3 \\ 1 & 4 \end{vmatrix} = (3 \cdot 4) - (3 \cdot 1) = 12 - 3 = 9$$

$$M_{12} = \begin{vmatrix} 1 & 3 \\ 3 & 4 \end{vmatrix} = (1 \cdot 4) - (3 \cdot 3) = (4 - 9) = -5$$

$$M_{13} = \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} = (1 \cdot 1) - (3 \cdot 3) = 1 - 9 = -8$$

$$M_{21} = \begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} = (3 \cdot 4) - (2 \cdot 1) = (12 - 2) = 10$$

$$M_{22} = \begin{vmatrix} 2 & 2 \\ 3 & 4 \end{vmatrix} = (2 \cdot 4) - (3 \cdot 2) = 8 - 6 = 2$$

$$M_{23} = \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix} = (2 \cdot 1) - (3 \cdot 3) = (2 - 9) = -7$$

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$M_{31} = \begin{vmatrix} 3 & 2 \\ 3 & 3 \end{vmatrix}$ $M_{32} = \begin{vmatrix} 2 & 2 \\ 1 & 3 \end{vmatrix}$ $M_{33} = \begin{vmatrix} 2 & 3 \\ 1 & 3 \end{vmatrix}$

$= (3\cdot3)-(3\cdot2)$ $= (2\cdot3)-(2\cdot1)$ $= (2\cdot3)-(1\cdot3)$

$= 9-6$ $= 6-2$ $= 6-3$

$= 3$ $= 4$ $= 3$

· Tabel minor matriks A

M_{ij}	Minor
M_{11}	9
M_{12}	-5
M_{13}	-8
M_{21}	10
M_{22}	2
M_{23}	-7
M_{31}	3
M_{32}	4
M_{33}	3

c. Susun matriks kofaktor C, matriks $adj(A) = C^T$. Hitung $A^{-1} = \frac{1}{det(A)} \cdot adj(A)$

Jawab :

$C_{11} = 9$ $C_{12} = -1(-5) = 5$ $C_{13} = -8$

$C_{21} = -1(10) = -10$ $C_{22} = 2$ $C_{23} = -1(-7) = 7$

$C_{31} = 3$ $C_{32} = -1(4) = -4$ $C_{33} = 3$

kofaktor $C = \begin{bmatrix} 9 & 5 & -8 \\ -10 & 2 & 7 \\ 3 & -4 & 3 \end{bmatrix}$ //

$Adj(A) = C^T$

$= \begin{bmatrix} 9 & -10 & 3 \\ 5 & 2 & -4 \\ -8 & 7 & 3 \end{bmatrix}$ //

$A^{-1} = \frac{1}{17} \cdot \begin{bmatrix} 9 & -10 & 3 \\ 5 & 2 & -4 \\ -8 & 7 & 3 \end{bmatrix}$

$= \begin{bmatrix} 9/17 & -10/17 & 3/17 \\ 5/17 & 2/17 & -4/17 \\ -8/17 & 7/17 & 3/17 \end{bmatrix}$ //

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e. Blok piksel asli $P = \begin{bmatrix} 120 & (50+N) & 200 \\ 80 & 9+2 & (10+N) \\ (30+N) & 90 & 170 \end{bmatrix}$

Lakukan enkripsi $E = A \cdot P$, dekripsi $P' = A^{-1} \cdot E$. Buktikan $P' = P$

Apa yg terjadi jika $\det(A) = 0$?

Jawab: $a = 1$, $N = 13$

$$P = \begin{bmatrix} 120 & 50+13 & 200 \\ 80 & 1+2 & 10+13 \\ 30+13 & 90 & 170 \end{bmatrix} = \begin{bmatrix} 120 & 63 & 200 \\ 80 & 3 & 23 \\ 43 & 90 & 170 \end{bmatrix} //$$

Enkripsi:

$$E = A \cdot P$$

$$= \begin{bmatrix} 2 & 3 & 2 \\ 1 & 3 & 3 \\ 3 & 1 & 4 \end{bmatrix} \cdot \begin{bmatrix} 120 & 63 & 200 \\ 80 & 3 & 23 \\ 43 & 90 & 170 \end{bmatrix}$$

$$= \begin{bmatrix} (2\cdot120)+(3\cdot80)+(2\cdot43) & (2\cdot63)+(3\cdot3)+(2\cdot90) & (2\cdot200)+(3\cdot23)+(2\cdot170) \\ (1\cdot120)+(3\cdot80)+(3\cdot43) & (1\cdot63)+(3\cdot3)+(3\cdot90) & (1\cdot200)+(3\cdot23)+(3\cdot170) \\ (3\cdot120)+(1\cdot80)+(4\cdot43) & (3\cdot63)+(1\cdot3)+(4\cdot90) & (3\cdot200)+(1\cdot23)+(4\cdot170) \end{bmatrix}$$

$$= \begin{bmatrix} 240+240+86 & 126+9+180 & 400+69+340 \\ 120+240+129 & 63+9+270 & 200+69+510 \\ 360+80+172 & 189+3+360 & 600+23+680 \end{bmatrix}$$

$$= \begin{bmatrix} 566 & 315 & 809 \\ 489 & 342 & 779 \\ 612 & 552 & 1303 \end{bmatrix} //$$

Dekripsi:

$$P' = A^{-1} \cdot E$$

$$= \begin{bmatrix} 9/17 & -10/17 & 3/17 \\ 5/17 & 2/17 & -9/17 \\ -8/17 & 7/17 & 3/17 \end{bmatrix} \cdot \begin{bmatrix} 566 & 315 & 809 \\ 489 & 342 & 779 \\ 612 & 552 & 1303 \end{bmatrix}$$

$$= \begin{bmatrix} 5094/17 + (-4890/17) + 1836/17 & 2835/17 + (-3420/17) + 1656/17 & 7281/17 + (-7790/17) + 3909/17 \\ 2830/17 + 978/17 + (-2448/17) & 1575/17 + 684/17 + (-2208/17) & 4045/17 + 1558/17 + (-5212/17) \\ -4528/17 + 3423/17 + 1836/17 & -2520/17 + 2394/17 + 1656/17 & -6472/17 + 5453/17 + 3909/17 \end{bmatrix}$$

$$= \begin{bmatrix} 2040/17 & 1091/17 & 3400/17 \\ 1360/17 & 51/17 & 391/17 \\ 731/17 & 1530/17 & 2890/17 \end{bmatrix} = \begin{bmatrix} 120 & 63 & 200 \\ 80 & 3 & 23 \\ 43 & 90 & 170 \end{bmatrix} //$$

terbukti $P' = P$ //

Studee 36 lines (6mm spaced)

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Jika det (A) = 0, maka matriks A merupakan matriks singular dan tidak memiliki invers. Jika matriks A tidak memiliki invers, maka proses perhitungan dekripsi untuk mencari nilai matriks P' tidak dapat dilakukan & tidak bisa dilakukan pembuktian bahwa P' = P.

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